$y[1]$ $y[2]$ $y[3]$ Measurements

$y[t] + a_1 y[t-1] + a_2 y[t-2] = 0$ (Plus initial conditions)

How do we find a_1, a_2?

$-y[3] = +a_1 y[2] + a_2 y[1]$

$$\begin{bmatrix} y[2] & y[1] \\ y[3] & y[2] \\ \vdots & \vdots \\ y[n-1] & y[n-2] \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} -y[3] \\ -y[4] \\ \vdots \\ y[n] \end{bmatrix}$$

$Ax = y$ Least squares solution

$\underline{x} = (A^T A)^{-1} A^T y$

Now that we have a_1, a_2 we can determine the characteristic equation

If real then $\gamma^2 + a_1 \gamma + a_2 = 0$

$y[t] = c_1 \gamma_1^t + c_2 \gamma_2^t$

Initial conditions

Real

$y[t] = c\gamma^t + c^* \gamma^t$ let $\gamma = \alpha + j\beta = re^{j\theta}$

$= c(\alpha + j\beta)^t + c^*(\alpha - j\beta)^t$

$= c(re^{j\theta})^t + c(re^{-j\theta})^t$

$$c r^t e^{i\theta} + c r e^{-j\theta}$$

$$|c| e^{i\theta} r^t e^{j\theta t} + |c| r^t e^{-j\theta} e^{-j\theta t}$$

$$2 r^t |c| \left[\frac{e^{j(\theta t + \phi)} + e^{-j(\theta t + \phi)}}{2} \right]$$

$$= 2 r^t |c| \cos(\theta t + \phi) \quad //$$

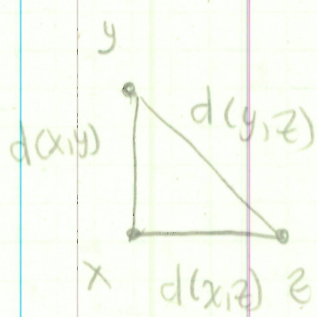
//

<u>Def</u> Let X be set. A metric

$$d: X \times X \to \mathbb{R}$$

is a function used to measure distance between elements in a set with the following properties

1) $d(x,y) = d(y,x)$
2) $d(x,y) \geq 0$
3) $d(x,y) = 0$ iff $x = y$
4) $d(x,z) \leq d(x,y) + d(y,z)$ ← Triangle inequality



Example

$X = \mathbb{R}n \quad \underline{x} \in \mathbb{R}^n \quad \underline{y} \in \mathbb{R}^n$

$$d_1(\underline{x}, \underline{y}) = \sum_{i=1}^{n} |x_i - y_i|$$ L_1 metric
Manhatten Metric

$$d_2(\underline{x}, \underline{y}) = \left(\sum_{i=1}^{n} (x_i - y_i)^2 \right)^{1/2}$$ L_2 Metric
Euclidean Metric

$$d_p(\underline{x}, \underline{y}) = \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p}$$ l_p Metric

$$d_\infty(\underline{x}, \underline{y}) = \max_{i \in 1, \dots, n} |x_i - y_i|$$ l_∞ metric

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